COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.3

This worksheet contains a combination of *formative activities* (which contribute towards your learning) and *summative activities* (which you will complete and submit to be assessed as part of your portfolio).

- Exercises marked with red borders are summative and must be submitted in Gradescope.
- Other exercises may be covered in tutorial sessions.
- Exercises marked by (*) are advanced, and may take some time to complete.
- Exercises marked by (*E*) are exam-style questions. You should expect to see questions like this in the exam.

Expectations:

1. **Timeliness** You should complete all of the activities in the order provided and submit your portfolio evidence on PebblePad before the deadline.
2. **Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff.
3. **Integrity:** You are responsible for ensuring that the evidence you submit as part of your portfolio is **entirely your own work**. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in tutorials and the Math Lunch Hour. Individual support is available via office hours.

Expected completion date: Friday 23 October 2025

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment. Within this assessment you may use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries.
- test your knowledge against a piece of content.
- help identify and correct spelling mistakes and grammatical errors in your work.
- provide feedback and advice on your overall coding style.

You must not use Gen AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning outcomes

On completion of this activity sheet you will have:

1. demonstrated an understanding of vocabulary relating to proof, including antecedent, conclusion, assumption, axiom, theorem, lemma, corollary, proof, conjecture, contradiction, contrapositive, induction, counterexample,
2. constructed formal mathematical proofs using direct proof, contradiction, and contrapositive.
3. applied mathematical induction to prove statements about positive integers and recursive algorithms,
4. applied inductive strategies to solve problems in computer science,
5. demonstrated an understanding of the basic terminology of set theory, including set, union, intersection, symmetric difference, complement, tuples, ordered pair, and Cartesian product.

Tutorials on Wednesday and Thursday will cover material from Week 3, Lecture 1. This material is in exercises 1–4. Tutorials on Monday and Tuesday will cover material from Week 3, Lecture 2. This material is in exercises 6–11.

Questions

1. Let a and b be positive integers. We want to prove that if a and b are odd, then $a + b$ is even.
   (a) Is there a best way to prove this? Direct proof, contrapositive proof, or proof by contradiction?
   (b) Can you give a proof (using some method)?
2. Let a, b, and p be positive integers, where p is prime. Imagine that we want to prove that if p divides ab, then either p divides a or p divides b.
   (a) If you wrote a direct proof, how would you start the proof? How would you end it?
   (b) If you wrote a contrapositive proof, how would you start the proof? How would you end it?
   (c) If you wrote a proof by contradiction, how would you start the proof? How would you end it?

   (Note that you do not need to provide a proof.)
3. Let $x_1, x_2, \ldots, x_n$ be a sequence of positive integers. We want to prove that if $x_1 + x_2 + \cdots + x_n = k$, then at least one of the numbers in $x_1, x_2, \ldots, x_n$ is greater than or equal to k/n.
   (a) Is there a best way to prove this? Direct proof, contrapositive proof, or proof by contradiction?
   (b) Can you give a proof (using some method)?

4. We want to write a proof of the following fact: if n is a positive integer and n^2 is odd, then n is odd.

   (a) Construct a proof by completing the following statements.

   (1) Assume that n is even.

   (2) Because _____, it can be expressed as _____, where k is some integer

   (3) Therefore n^2 is _____.

   (4) Because _____, it follows that n^2 is even.

   (b) What style of proof have we just used to show that if n^2 is odd then n is odd?

5. (*E*) Give an inductive proof that

$$1 + 2 + 4 + \cdots + 2^n = 2^{n+1} - 1$$

   for every positive integer n. Make sure you check the base case, state the inductive hypothesis, and explain when you are using the inductive hypothesis.

6. (*E*) Give an inductive proof that

$$2 + 4 + 6 + \cdots + 2n = n^2 + n$$

   for every positive integer n. Make sure you check the base case, state the inductive hypothesis, and explain when you are using the inductive hypothesis.

7. (*E*) Remember that a number p is divisible by 8 if it can be expressed as $p = 8q$, where q is some integer.

   Use induction on n to prove that $(2n + 1)^2 - 1$ is divisible by 8 for every positive integer n. Make sure you check the base case, state the inductive hypothesis, and explain when you are using the inductive hypothesis.

8. Let n be an integer satisfying $n \geq 3$. Use induction to prove that a polygon with n corners can be covered by $n - 2$ triangles in such a way that the triangles do not overlap, and the corners of each triangle line up with corners of the original polygon. (Assume that the polygon is *convex*, meaning that every corner has an internal angle less than $180°$.)

9. Remember that a number p is divisible by 5 if it can be expressed as $p = 5q$, where q is some integer.
   Use induction on n to prove that $8^n - 3^n$ is divisible by 5 for every positive integer n.
   Make sure you check the base case, state the inductive hypothesis, and explain when you are using the inductive hypothesis.

10. (*E*) Let A, B, and C be sets. Draw Venn diagrams that represent each of the following sets. (Remember that you need to include the universal set in your Venn diagrams.)

    (a) $\overline{(A \cup B) \cap (B \cup \overline{C})}$

    (b) $A \triangle (\overline{B \cap \overline{C}})$

    (c) $(A \cap B) \setminus \overline{C}$

11. (*E*) Using the set identities in Proposition 5.12 of the lecture notes, prove that $\overline{\overline{A} \cup (B \cap \overline{A})}$ is equal to $A \cap (A \cup \overline{B})$. Explicitly state which identity you are using at each step.

    Why is $A \cap (A \cup \overline{B})$ equal to A?

12. Assume that U is the universal set. Why is $B \cup \overline{B} = U$ true? Why is $A \cap U = A$ true?
    Using these facts and the identities in Proposition 5.12, prove that $(A \cap B) \cup (A \cap \overline{B}) = A$.

13. Let A be the set $\{1, 3, 5\}$ and let B be $\{2, 4, 6\}$. List the elements of the set

$$\{X \in \mathcal{P}(A \cup B) \mid X \cap A \text{ has the same number of elements as } X \cap B\}.$$

14. What are the elements of $\mathcal{P}(\{\emptyset, \{\emptyset\}\})$?

15. Let A, B, and C be sets. Consider the following sets.

    (i) $(\overline{A} \cap \overline{B}) \cup (B \cap (A \cup C))$ (ii) $((\overline{A \triangle B}) \setminus C) \cup (A \cap C)$ (iii) $(C \setminus (A \setminus B)) \cup \overline{A \triangle B}$

    (a) For each of the three expressions draw a Venn diagram with three circles and shade the region corresponding to that expression.

    (b) Which two expressions represent the same set?

16. (*) Assume we have an algorithm which has a running time equal to $T(n)$ when the input has size n. We know that $T(1) = 3$ and

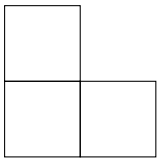
$$T(n+1) = 2T(n) - n + 1$$

    when n is a positive integer.

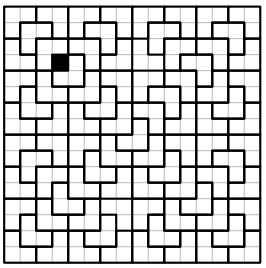
    (a) By calculating values of $T(n)$, can you guess a formula for $T(n)$?

    (b) Are you able to use induction to prove that your formula is correct for every value of n?

17. (*) A *tromino* is a tile made up of three squares, as in the picture below.



    Let n be a positive integer. Prove by induction on n that if we take a $2^n \times 2^n$ grid of squares, and remove any one of these squares, then we can cover the remaining squares by tromino-shaped tiles in such a way that no tiles overlap. (Note that we are allowed to rotate the tiles.)



18. (*)

(a) Express 1, 3, 5, and 7 in the form $a^2 - b^2$, where a and b are non-negative integers.

(b) From this experiment, can you see a way to directly prove that every odd positive integer can be expressed as $a^2 - b^2$, where a and b are non-negative integers?

19. (*) Let A be a set and let k be a positive integer. A *partition* of A into k *parts* is a collection of k non-empty subsets of A, such that every element of A is in exactly one of these subsets. For example, if A is $\{1, 2, 3, 4, 5, 6\}$, then $\{1, 3\}$, $\{2\}$, $\{4, 5, 6\}$ is a partition of A into 3 parts.

(a) Let A be $\{1, 2, 3, 4\}$. Find all partitions of A into 2 parts, and all partitions into 3 parts.

(b) If n is a positive integer and k is an integer satisfying $1 \leq k \leq n$, then $S(n, k)$ stands for the numbers of partitions into k parts of a set with n elements. This is called a *Stirling number of the second kind*. Prove that $S(n + 1, k) = kS(n, k) + S(n, k - 1)$. (This fact was first discovered in 1782 by the Japanese mathematician Masanobu Saka.)